

$$(1)(\text{最大值}) = 2a - 3 \quad (x = 3) \qquad (2)(a \geq 0) a \leq x \leq 2a, \quad (a \leq 0) 2a \leq x \leq a \qquad (3) 0 < a < \frac{3}{2}, \quad 6 < a$$

$$(1)0 \qquad (2)P(x)=(x-1)(x^2-2ax-a+6), \qquad a=\frac{7}{3}, \, x=\frac{11}{3} \qquad (3)2\leq a\leq \frac{7}{3}$$

$$(1) 5 + 2\sqrt{6} \quad (2) 2\sqrt{6} - 4, \quad b^2 + 8b = 8 \quad (3) a^2 - b^2 - a + b = 36\sqrt{6}, \quad (b+2)(b+3)(b+5)(b+6) = 460$$

$$(1) \frac{1}{15}, \quad \frac{8}{15} \qquad (2) \frac{1}{225}, \quad \frac{4}{75} \qquad (3) \frac{12}{49}$$

$$1-3 < x < 3 \qquad (2)1 \leq a \qquad [2](1)(\mathcal{A})-\frac{\sqrt{2}}{2} \qquad (\mathcal{I})5 \qquad (2)(i)1 \qquad (ii)\frac{2}{3}$$

$$(1)a = 6, \quad x = 1, 2, 3 \qquad (2)k \leq 6 - 4\sqrt{2}, 6 + 4\sqrt{2} \leq k \qquad (3)k = -2, \quad x = 2, 3 \pm 2\sqrt{2}$$

$$(1) M(2, 2), AB = 4\sqrt{2} \quad (2) C : (x-2)^2 + (y-2)^2 = 8, DE = 2\sqrt{6} \quad (3) (x-4)^2 + (y-2)^2 = 2, \quad 6\sqrt{3}$$

$$(1) \sin \theta \cos \theta = \frac{1-t^2}{2} \quad (2) t = \sqrt{2} \sin \left(\theta - \frac{\pi}{4} \right), \quad -\sqrt{2} \leq t \leq \sqrt{2} \quad (3) 1 \leq y \leq 4 + 2\sqrt{2}, \quad 1 < a < 4 - 2\sqrt{2}$$

$$(1)a = 3 \quad (2)(\text{極大値})6, \quad (\text{極小値})5, \quad x = 1, \frac{5}{2} \quad (3)k = \frac{1}{2}, \quad 2 + \sqrt{5}$$

$$(1)a_n = 4n - 3 \qquad (2)b_n = 2^n \qquad (3)N = 25, \quad \sum_{k=1}^N d_k = 1641$$

$$(1) \overrightarrow{OE} = \frac{2}{3} \overrightarrow{a}, \overrightarrow{BE} = \frac{2}{3} \overrightarrow{a} - \overrightarrow{b} \quad (2) \overrightarrow{OP} = \frac{2}{3} s \overrightarrow{a} + (1-s) \overrightarrow{b}, s = \frac{5}{6} \quad (3) \overrightarrow{PH} = -\frac{5}{9} \overrightarrow{a} + \frac{5}{18} \overrightarrow{b}, |\overrightarrow{PH}| = \frac{10\sqrt{2}}{9}$$

$$(1) \cos \alpha = \frac{\sqrt{6}}{3}, \sin 2\alpha = -\frac{2\sqrt{2}}{3} \quad (2) y = 2 \sin \theta + 2\sqrt{2} \cos \theta \quad (3) (\text{最大值}) 2\sqrt{3}, (\text{最小值}) -2\sqrt{2}$$

$$(1)(\mathcal{A})_{\frac{1}{2}}, \quad (\mathcal{I})-2, \quad (\mathcal{U})0, \frac{5\pi}{6}, \pi \qquad (2)b=1, \quad \theta=0, \pi, \frac{4\pi}{3}$$